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Tracing Subsurface Flow on Roadcuts on Steep, Forested Slopes¹

WALTER F. MEGAHAN AND JAMES L. CLAYTON²

ABSTRACT

In situ hydraulic conductivity was determined in a forested soil in the Idaho batholith by a tracer technique, and values were compared to hydraulic conductivity determined in the laboratory on cored samples. We injected NaCl solutions into piezometers placed in soil above a roadcut, and determined time for Na^+ or Cl^- to appear in outflow at the roadcut during saturated subsurface flow generated from spring snowmelt. Hydraulic conductivity values determined with the NaCl tracer averaged an order of magnitude greater than laboratory derived values. This is likely due to rapid flow in macropores formed from root channels or other biological activity that cannot be sampled by conventional coring and laboratory conductivity tests. Although the field tests indicated flow was clearly non homogeneous and anisotropic, an evaluation of the Reynold's number for the test conditions indicated Darcy flow was occurring. Sodium analysis provided a more distinct peak than Cl^- when both concentrations were plotted vs. time.

Additional Index Words: saturated hydraulic conductivity, macropore flow, granitic soil.

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SUBSURFACE FLOW is defined by Chow (1964) as “that part of the precipitation which infiltrates the surface soil and moves laterally through the upper soil horizons toward the stream as ephemeral, shallow, perched groundwater above the main groundwater level.” He suggests that flow velocities tend to be rapid with the comment that “a part of the subsurface flow may reach the stream promptly.”

Various workers suggest that subsurface flow velocities are particularly high on forested slopes because of the presence of organic accumulations, root channels, earthworm holes, etc. (Hursh and Hoover, 1941;

Fletcher, 1952; Rothacher et al., 1967; Mosley, 1979). There has been considerable interest in studies designed to measure the amount of subsurface flow occurring on forested slopes (Whipkey, 1965; Takeda, 1967; Harr, 1977; Mosley, 1982). These studies have raised some questions regarding the contribution of subsurface flow to the storm hydrograph as contrasted to overland flow generated according to the partial area concept (Dunne and Black, 1970; Stephenson and Freeze, 1974; Mosley, 1979). The saturated hydraulic conductivity (K) of the subsurface flow zone becomes an important factor in determining the velocity of flow through the slope.

Whipkey (1965) noted that the following conditions are particularly conducive to subsurface flow: (i) the land is sloping, (ii) surface soil is permeable, (iii) a water-impeding layer is near the surface, and (iv) the soil is saturated. Based on these criteria, conditions are ideal for subsurface flow in the Idaho batholith, a massive intrusion of granitic rocks covering approximately 40 000 km² in Idaho and part of Montana. Typically, shallow, coarse-textured soils (loamy sands and sandy loams) are found on steep slopes that average 60% or more in many drainages. The granitic bedrock exhibits various degrees of weathering and fracturing; however, saturated hydraulic conductivities are almost always less for the bedrock than for the overlying soil. Thus, the bedrock usually impedes the downward flow of water. Cyclonic storms and especially deep snow packs that release large volumes of water to the soil within short periods are annual occurrences in this area. Water supplied by these climatic events rapidly infiltrates the soil surface and continues to flow downward until it reaches a zone of reduced vertical hydraulic conductivity at the bedrock surface. Continued infiltration creates a saturated layer at the soil-bedrock interface, causing subsurface flow along the bedrock surface.

Road construction in such mountainous terrain necessitates cutting into the hillside to provide a level roadbed. This commonly incises bedrock on moderate

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² Principal Research Hydrologist and Research Soil Scientist, respectively, Intermountain Station's Forestry Sciences Laboratory, Boise, Idaho.

to steep slopes. As a consequence, a large portion of the subsurface flow that occurred in the slope before construction is intercepted by the cut slope of the road leading to accelerated erosion and sedimentation problems. In 1969, an intensive subsurface flow study was initiated on two small watersheds to evaluate the effects of logging on the rate and volume of subsurface flow intercepted by logging roads (Megahan, 1983). The objective of the study reported here is to evaluate saturated hydraulic conductivity with the use of tracers during active subsurface flow periods.

MATERIALS AND METHODS

Study Area

The study area is in the Pine Creek drainage, a tributary of the Middle Fork of the Payette River located in west-central Idaho. The area consists of a first-order watershed of approximately 1 ha. This microwatershed is representative of undisturbed, first-order drainages found in the mid-elevation (1500–1600 m), forested landscapes of the Idaho batholith. Slopes average 55% and face north. Vegetation consists of an overstory of ponderosa pine (*Pinus ponderosa* Laws.), lesser amounts of Douglas-fir (*Pseudotsuga menziesii* mirb. (Franco) and Englemann spruce (*Picea engelmannii* Parry), and an understory of small trees and shrubs. The soil is classified as Koppes loamy coarse sand and is a member of the Sandy-Skeletal Mixed family of Typic Cryoborolls (Nelson, 1976). The soils are underlain by moderately weathered and fractured, granitic bedrock. Soil depths average 0.65 m. These soils are weakly developed, exhibiting only thin A and C horizons. The saturated hydraulic conductivity of the underlying bedrock averages about 1 to 2 orders of magnitude less than the saturated hydraulic conductivity of the soil (Hampton et al., 1974) causing a buildup of water and subsurface flow along the soil-bedrock interface.

Annual precipitation in the study area averages approximately 0.80 m. A large portion occurs as snowfall; consequently, snowpacks in excess of 1.50 m (0.50 m of water equivalent) often accumulate. Rapid spring snowmelt contributes large volumes of water to the soil generating subsurface flow.

Data Collection

An in-sloped logging road marks the lower boundary of the study watershed. Roadcuts are deep enough to expose bedrock across the entire length of both watersheds. During periods of high infiltration, subsurface flow emerges on the face of the roadcut at the soil-bedrock interface and runs down the granitic bedrock (Fig. 1).

Piezometers were constructed and installed using the procedures described by Swanston (1967). Piezometer access holes were drilled to a depth of at least 60 mm in the underlying bedrock. Piezometers were placed into the hole and the hole was backfilled with 2- to 3-mm size sand up to a depth of 300 mm from the soil surface. The remainder of the hole was backfilled with bentonite clay to prevent inflow of water at the soil surface. Depth of the phreatic surface was measured to ± 2.0 mm during the tracer tests. Soil samples taken during the course of borehole shear tests in the phreatic zone were used for particle size analysis (Gray and Megahan, 1981).

Generally, tracer studies are carried out by injecting tracer into an upslope piezometer and sampling one or more downslope piezometers. However, we were concerned with the possibility that actual flow paths might be influenced by such factors as root channels and irregularities in the bedrock surface to the point that downslope piezometers might

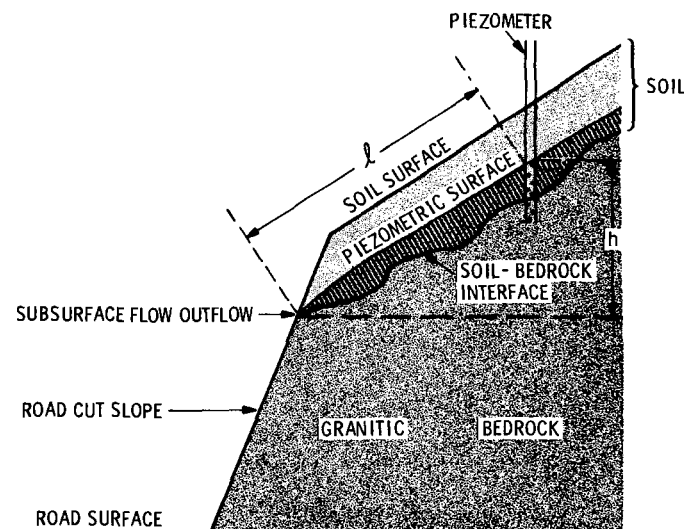


Fig. 1—Diagram showing piezometer location, soil-bedrock contact, and free-water surface in a slope above a roadcut.

not be in a position to accurately sample flow velocities. Accordingly, we injected tracers in piezometers located directly above the roadcut, allowing us to find the downslope outflow point that is hydraulically most efficient by sampling anywhere along the roadcut face rather than being confined to the piezometer sites. Piezometer no. 12, 14, and 9, representing slope distances of about 1.5, 3, and 6 m, respectively, from the piezometers to the roadcut were selected for study because of their close proximity to the roadcut. Hereafter, piezometer no. 12, 14, and 9 will be referred to as 1.5, 3, and 6 to correspond to the approximate flow length. The soil depth (depth to restricting layer) sampled for each piezometer was 0.74, 0.58, and 0.84 m for piezometers 1.5, 3, and 6, respectively.

Sodium chloride was used as a tracer for this study. Selection of this particular tracer was based on cost, ease of use, and the fact that the granitic soils are low in exchange surfaces so that adsorption and dispersion problems are minimized. Flow velocity tests were conducted by adding a slug of 6M NaCl solution to the test piezometer at time 0. Successful tests were obtained using 200 mL of tracer in piezometers 1.5 and 3 and 500 mL of tracer in piezometer 6. The point of first emergence on the roadcut was located by sampling at 300-mm intervals along the face of the roadcut every 10 min. Sample sites were centered at the most probable point of emergence (the apparent topographic outflow point) and extended laterally on both sides of this point for a distance equal to at least one half the distance from the piezometer to the roadcut. Tracer was detected by dropping 0.1M AgNO₃ into a 50-mL water sample and observing for a precipitate of white AgCl. As soon as the point of first emergence was located, a small collection trough was inserted at the point to facilitate sampling (Fig. 2). Fifty-milliliter water samples were collected at this location at 10- to 15-min intervals until there was a definite reduction in tracer concentration as evidenced by a reduction in the amount of AgCl precipitate. Water samples were transported to the laboratory where they were analyzed for both Na and Cl by flame emission spectroscopy and titration with AgNO₃ (argentometric method, Taras et al., 1971), respectively.

A total of seven tracer tests were conducted during the period from 10 to 13 April 1971, two in piezometer 1.5, two in piezometer 3, and three in piezometer 6. Water temperature at the outflow points was taken periodically during each test. In addition, each test site was surveyed to determine the flow length (l) and the differences in elevation (h) between the phreatic surface in each piezometer and the

outflow point (Fig. 1). Straight line flow paths were assumed between the piezometer and the point of first emergence. Following the period of active subsurface flow, soil pits were dug along the assumed flow path to the depth of the bedrock restricting zone. Soil cores were taken from the sides of the pits in the estimated center of the subsurface flow zone with the vertical axis of each core oriented parallel to the bedrock surface and in the direction of flow. Cylindrical cores 54 mm in diam and 60 mm long were obtained using a commercially available core sampler. A total of eight core samples were obtained, three each representing piezometers 1.5 and 3 and two representing piezometer 6.

Soil cores were carefully transported to the laboratory where saturated hydraulic conductivity tests were run directly on the undisturbed soil cores using a falling head permeameter (Todd, 1959). An average of five tests were used to determine the K values for each soil core. Samples were then oven-dried at 378K for a 2-d period, volume weight was determined and porosity computed, assuming a solid particle density of 2.65 Mg m^{-3} . Samples were then carefully inspected before and after removal from the sample cylinders to assure good contact between cores and cylinders; one sample from piezometer 3 contained obvious voids caused by mishandling and was discarded.

Field estimates of hydraulic conductivity, K , for the tracer studies were computed in the following manner. Pore velocity for each tracer test, v_p , was calculated by dividing straight line flow path in millimeters, by time to peak concentration of tracer in seconds. Since flow is restricted to pores, velocity of flow through a cross sectional area of soil was calculated by adjusting for porosity (Todd, 1959):

$$v = \alpha V_p, \quad [1]$$

where v is Darcy velocity in millimeters per second and α is porosity. Values of α were determined from the core samples of the flow path for each piezometer. Finally, hydraulic conductivity, K , was computed from the Darcy equation using Darcy velocity:

$$K = v dl/dh. \quad [2]$$

where l is distance in meters from the level of the phreatic surface at the piezometer to the outflow point and h is the difference in elevation in meters between the phreatic surface in the piezometer and the outflow point at the time of test.

The standard falling head permeameter equation was used to calculate laboratory K values (Todd, 1959).

$$K = [(d_t^2 \cdot lc)/d_c^2 \cdot t] \ln (h_0/h_t), \quad [3]$$

where d_t is the diameter of water delivery tube in mm, d_c is the diameter of soil core in mm, l_c is the length of soil core in mm, t is the elapsed time of test in s, h_0 is the initial head of water in mm, and h_t is the head of water at time t in mm.

All K values were adjusted to a standard temperature of 293°K to account for differences in water viscosity for the field and laboratory tests as follows:

$$K_{293} = K_{\eta} \phi_{293} / \eta_{293} \phi_t, \quad [4]$$

where K is hydraulic conductivity in mm s^{-1} , η is kinematic viscosity in $\text{mm}^2 \text{ s}^{-1}$, ϕ is water density in Mg m^{-3} , 293 refers to temperature, $^\circ\text{K}$ and t is temperature in $^\circ\text{K}$ at the time of test.

RESULTS AND DISCUSSION

A total of seven tracer tests were run during the study: two on piezometer 1.5, two on piezometer 3, and three on piezometer 6. We were able to obtain good definitions of tracer concentration over time for both tests on piezometer 1.5, one of the two tests on piezometer 3, and one of the three tests on piezometer

Table 1—Details of successful tracer tests.

Data items	Sample no.			
	1.5-1	1.5-2	3	6
Water depth in piezometer, mm†	73	30	30	122
Flow path length, m	1.45	1.41	2.94	6.32
Total head difference, m	0.87	0.81	1.74	3.00
Hydraulic gradient	0.60	0.58	0.59	0.47
Porosity	0.40	0.40	0.35	0.41
Time to start of rise Na^+ , min	110	72	160	643
Time to start of rise Cl^- , min	101	72	160	540
Time to peak concentration Na^+ , min	180	202	325	1050
Time to peak concentration Cl^- , min	160	214	311	1060

† Water depths represent the vertical depth from the top of the restricting layer.

6. Problems were magnified as the flow lengths became longer because of the trial and error required to select the correct tracer dosage and the longer time needed for sample collection. Details of the successful tracer tests are summarized in Table 1.

Flow emergence along the roadcut on the soil-bedrock interface was almost continuous except for localized concentrations. As might be expected, the point of first emergence of the tracer tended to occur at points of localized flow concentration (Fig. 2). In tracer tests for two out of the three piezometers, however, the point of first detection was not the most direct flow path based on surface topography. For example, the point of first detection on piezometer 3 was 20° to the left of the path of most direct flow, if we assume that the medium was homogeneous and isotropic, and the flow path followed the gradient of the surface topography. In addition, the tracer was often detected emerging at two points that were separated by outflow containing no tracer. These results illustrate that hy-



Fig. 2—Sample location for tracer test on piezometer 1.5. Flow is concentrated in a zone of high root density associated with an argillized pegmatite dike. The metal trough is placed in this zone to facilitate sampling.

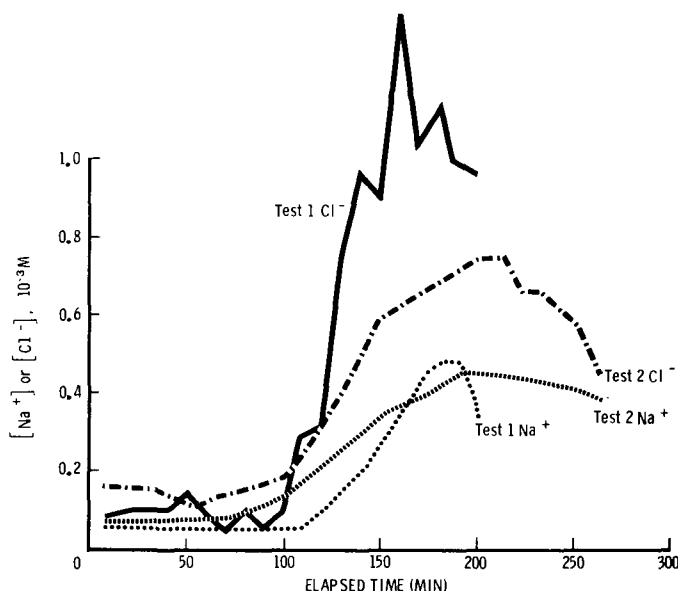


Fig. 3—Plots of Na^+ or Cl^- concentration vs. time for field tracer tests for piezometer 1.5.

draulic heterogeneity is probably the rule rather than the exception on forested, granitic slopes in this area.

Concentrations of both Na^+ and Cl^- were plotted over time since injection to provide a measure of flow velocity based on the time to the start of rise of concentration and the time to peak concentration. The time distribution for tracer was well defined for Na^+ for all successful tests and the time of peak outflow and especially the time of start of rise, were easily determined. Because of less precise analysis techniques, similar plots for Cl^- were less satisfactory, showing more scatter and poorer definition of start of rise and peaks (Fig. 3). In two out of eight cases, no differences were formed between the time to start of rise and time to peak for Na^+ compared with Cl^- (Table 1). In the rest of the tests, differences were small except for piezometer 6, where the time to start of rise was 19% greater for Na^+ compared with Cl^- . This large difference was a direct result of the scatter associated with the Cl^- analyses coupled with the attenuation of the concentration curve caused by the increased flow distance. There was a tendency for more rapid outflow of Cl^- compared with Na^+ , but the differences were not statistically significant at the 95% confidence level.

Hydraulic conductivity values were calculated for each tracer test, using velocity values determined for both Na^+ and Cl^- (Table 2). Differences in the K values for cation and anions are caused by the differences in the velocity measured for each ion. Again, there are no statistically significant (95% confidence level) differences between the means calculated for each ion.

Values average about two times greater when calculated using velocity determined by time to start of rise compared with time to peak. The decision of which value is appropriate will depend on the needs of the user. For our purposes, we feel the time to peak value is better because it is more representative of average flow conditions and the magnitude of errors introduced by diffusion processes is probably less.

Table 2—Computed hydraulic conductivities for field tracer and laboratory core tests.

Piezometer	Hydraulic conductivity, mm min^{-1}							
	Field tracer tests				Lab tests			
	Start of rise		Peak		Cores			
	Na^+	Cl^-	Na^+	Cl^-	1	2	3	Mean
1.5—1	15.2	16.0	9.3	10.5	0.57	0.70	0.34	0.54
1.5—2	13.6	23.6	8.4	7.9				
3	18.9	18.9	9.3	9.7	0.52	2.37	—	1.44
6	14.8	17.7	9.0	9.1	0.84	1.16	—	1.00
Avg	18.1	19.2	9.0	9.3				0.93

The choice of an anion or a cation as a tracer can be greatly influenced by processes of adsorption in some soils. The soils on the experimental watersheds have cation exchange capacities of 15 to 25 $\text{cmol} (\text{NH}_4^+) \text{ kg}^{-1}$ in the surface horizon and 5 to 8 $\text{cmol} (\text{NH}_4^+) \text{ kg}^{-1}$ in the C horizons. At the injection sites, the cation exchange capacities were $< 0.4 \text{ cmol} (\text{NH}_4^+) \text{ kg}^{-1}$. For this reason, we felt adsorption would have a negligible effect on the rate of cation travel. We did not analyze anion exchange capacities in these soils.

The saturated hydraulic conductivity values for the soil cores are also shown in Table 2. The lower mean K values for the tracer tests are based on time to peak velocities. Even though these are the lowest values for the tracer tests, they are still an order of magnitude greater than the average K values determined for the soil cores. The data suggest that tracer flow was occurring in macropores not sampled by the soil cores. Observations of the outflow zone on the roadcut surface indicated that both physical discontinuities in the parent material, such as fractures and zones of alteration, and root channels were primary pathways for flow. Figure 2 shows the point of first emergence of the tracer from piezometer 12, a high-density root zone located in the vicinity of a pegmatite dike in the bedrock.

The K value for test one on piezometer 1.5 was about 12% higher than the K value for test 2 at the same location. This was a direct result of the shorter time to peak for test 1 (Fig. 3). The phreatic surface was about 2.4 times higher during test 1 than it was during test 2 (7.3 vs. 3.0 cm) so that flow was occurring nearer the soil surface. This suggests more rapid flow through macropores possibly in response to greater biological activity near the surface. Other researchers have reported such effects on forest soils (Aubertin, 1971; Harr, 1977). Apparently, these soils are hydraulically heterogeneous in the vertical direction along with the lateral heterogeneity detected on the roadcut face.

Whipkey (1967) reported that subsurface flow measured on plots in Ohio was clearly non-Darcian as indicated by obvious turbulence in flow emerging from root channels. In view of this and the fact that we were sampling flow in a heterogeneous medium that was associated with roots and bedrock discontinuities, the question arises as to whether or not we were sampling Darcian flow conditions.

In general, saturated flow through porous media can be described by (Freeze and Cherry, 1979):

$$v = Ki^n, \quad [5]$$

where v = flow velocity, K = saturated hydraulic con-

ductivity, i = hydraulic gradient, and n = an exponent that varies with flow regime.

For Darcian flow, the exponent is assumed to be 1 and is a reflection of laminar flow conditions where viscous forces predominate. As flow velocities increase because of larger pores, inertial forces increase and some energy tends to be expended due to turbulence. Eventually, velocities become so great that the exponent drops below 1 and Darcy flow conditions no longer exist. Unfortunately, there are no absolute criteria for discriminating when n is < 1 . Guidelines do exist, however, based on the Reynold's number calculated as follows (Freeze and Cherry, 1979):

$$Re = \phi v d / \eta \quad [6]$$

where ϕ = mass density of the fluid, Mg m^{-3} , v = average flow velocity, m s^{-1} , d = mean diameter of the soil grains, m , and η = viscosity of the fluid, $\text{Mg m}^{-1} \text{s}^{-1}$.

Using the data from the tracer tests and a value for d of 0.001 mm determined from the particle size analysis, we calculated an Re value of about 0.1. Bear (1972) concluded that Darcy's law is valid as long as Re based on average grain diameter is not greater than some value between 1 and 10. Thus, we conclude that Darcian flow conditions existed at the time of our tests.

CONCLUSIONS

At best, detection of saturated subsurface flow in soils with piezometers is difficult because of variations in the physical properties of the soil and the transitory nature of subsurface flows (Betson et al., 1968). Nevertheless, subsurface flow is an important hydrologic component, especially on forested lands in mountainous areas, and information is needed to help better define subsurface flow processes. Tracer studies are a logical way to help accomplish this. Tracer detection between successive piezometers on a slope can be difficult, or at best, misleading, because of anisotropy and heterogeneity in the soil medium. Tracer detection at roadcuts, as described in this study, provides a solution to this problem because numerous outflow points can be sampled.

Hydraulic conductivity values determined by our tracer techniques were about an order of magnitude greater than values obtained in the laboratory from undisturbed cores. The large differences appeared to be caused by greater flow along macropores formed by roots and other biological activity and discontinuities in the parent material that were carrying flow in the tracer studies but were not sampled by coring. These results reinforce the need for field measurements to obtain estimates of actual K values.

The tracer technique used in this study is simple, fast, and cheap and allows for establishing whether or not anisotropy is present. Even though the flows were clearly nonhomogeneous and anisotropic, an evaluation of the Reynold's number for the test conditions suggest that Darcy flow was occurring during the tests. The Na cation rather than the Cl anion was best for the coarse textured soils in our study, possibly because of minimal adsorption potential and the fact that the detection technique for sodium was more precise. Tracers were detectable up to 6 m, but the tests at 1.5

and 3 m were easier because application rates and sampling times were more easily determined.

Some error is introduced by sampling at roadcuts compared with sampling between successive piezometers because the flow distance was not corrected for drawdown at the roadcut face. This error would be small for the shallow phreatic depths sampled, however, and is more than compensated by the ability to identify the point of first emergence at roadcuts. We recommend this procedure with the warning that adsorption problems be carefully evaluated before using any tracer.

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